Field-Specific Evidence Contracts for Dual-Arm Robot Feedback under Explicit Error Budgets

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Abstract

Reliable robot monitoring requires more than recent timestamps and finite feedback values. We present a field-specific evidence contract and evaluate it in dual-arm execution and feedback-sampling experiments. The contract specifies joint names, required fields, native units, producer validity and clock continuity before testing a trapezoidal position–velocity relation. Explicit position, velocity, timestamp and model-discrepancy allowances yield a necessary consistency bound and an exact fixed-clock minimum velocity adjustment. All 1,968 execution-feedback rows passed a 20 ms age check, yet one right-arm trace held its position–velocity vector for 0.664 s. In matched 0.5 s bins, this trace required a minimum velocity adjustment of 1.8976 degrees/s, compared with at most 0.01896 degrees/s in 311 direct sampling bins. With 0.01-degree endpoint and model allowances and 0.1 ms timestamp allowance, its adjustment lower bound remained 1.8352 degrees/s. A 1 ms timestamp allowance instead required abstention because sample order was not robust to the allowed perturbations. Independent linear programs agreed with the fixed-clock expression in 1,396 numerical problems. Supporting schema comparisons recovered 54,897 arm-eligible messages rejected by whole-message finiteness and withheld six finite zero-velocity rows marked unavailable by their producer. The practical gain is a quantitative, reusable distinction between calculable feedback, bounded inconsistency and insufficient evidence. The single sustained inconsistency and illustrative, uncalibrated error budgets do not establish physical fault-detection accuracy.

Keywords: Robot diagnostics, Data provenance, Runtime monitoring, Joint feedback, Bounded errors, Fault observability

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1 Introduction

Dual-arm robots use joint feedback to admit actions, monitor execution and respond to faults. Asynchronous sensing, estimation and communication layers can give different fields different update histories [1]. Consequently, a recent message with finite values can still provide contradictory or insufficient evidence about motion. Conversely, an invalid field outside the requested arm can cause a whole-message check to discard usable inputs. These errors matter because the monitor must decide whether to evaluate a motion relation, report a discrepancy, or withhold judgment.

Observability-based diagnosis combines logs, traces and trajectories to explain robotic failures

[2, 3]. Tracing and runtime verification expose execution events and evaluate explicit properties [4, 5]. Bounded-error diagnosis tests consistency with an uncertainty set [6]. These approaches motivate a narrower unresolved problem: how to connect the prerequisites of a feedback calculation to the strength of its conclusion. A timestamp check does not test position–velocity consistency, and a consistency residual alone does not establish that its required fields or error assumptions are valid.

The central research question is: *when joint feedback is the only motion signal, how can a monitor distinguish unavailable evidence from a position–velocity contradiction under explicit error bounds?* Three questions make this testable. **RQ1:** Which inconsistencies remain after age, speed and finite-value checks? **RQ2:** How much declared measurement and model error reconciles the eligible kinematics? **RQ3:** What useful inputs are retained, and when must the method return undetermined?

Three challenges determine the method. Joint names, missing arrays and producer flags require property-specific input validation. Shared clocks and feedback chains require separating internal consistency from physical accuracy. Uncertain timestamps and model discrepancy require a bounded relation with an explicit abstention condition. We address these challenges through a field-specific evidence contract that declares required joints, fields, units, validity and one integration clock before calculation. It then reports a minimum adjustment budget and a conditional verdict, or the reason a verdict is unavailable.

The objective is to make that contract executable and evaluate its practical gain over named monitoring predicates on the same inputs. Dual-arm execution and feedback-sampling experiments supply the measured comparisons; independent optimization checks the fixed-clock expression. The resulting method connects input eligibility, quantified inconsistency and justified abstention in one reproducible procedure. Its contribution is this connection and its measured consequences, without estimating physical fault-detection accuracy from the single sustained case.

2 Related work

2.1 Fault diagnosis and bounded-error reasoning

Robotic fault diagnosis uses model residuals, signal processing and learned representations; deployment constraints remain important across these approaches [7]. Cacace et al. combine external-wrench estimation with a recurrent model for motor-fault isolation in simulation [8]. Reppa and Tzes use set-membership identification and output bounds for a simulated two-link manipulator [6]. Those methods test specified fault or dynamic-model hypotheses. Here the available signals support a simpler discrete kinematic relation. We quantify the correction needed for that relation and make its input requirements explicit.

Song et al. demonstrate the value of combining robotic observability data [2], while ROBUST emphasizes reproducible software-bug cases [3]. Our contribution concerns a limitation shared by such diagnostic workflows: several fields or views can originate from the same uncertain acquisition chain. Numerical disagreement can be established without identifying which producer, estimator or communication layer caused it.

2.2 Monitoring tools and measurement semantics

ROS timestamp and frequency diagnostics assess communication timing [9]. Joint-state broadcaster supports joint/interface selection and field mapping [10]. Tracing exposes computation and communication events [4]. Ogma, ROSMonitoring 2.0 and RTAMT support explicit runtime properties and temporal evaluation [11–13]; Caldas et al. discuss field-testing conditions and oracle definition [5]. These capabilities motivate, rather than replace, the contract's field and error assumptions. The comparison gap concerns the premises and outputs of individual checks. The present study therefore evaluates named predicates on identical inputs; it does not benchmark complete competing software packages.

Three-valued runtime semantics already distinguish a justified verdict from insufficient observation [14]. We apply that distinction to field eligibility and bounded kinematic interpretation.

Age of information likewise requires a generation-time definition [15]; a hardware abstraction layer (HAL) stamp need not identify each field's device acquisition. Controller action success has its own tolerance and execution semantics [16]. Keeping these meanings separate follows the distinction among faults, erroneous states and service failures [17].

3 Experiments and methods

3.1 Dual-arm execution and feedback sampling

The sole author designed and performed the dual-arm operation and feedback-sampling experiments and collected the resulting records. The execution experiment took place on 26 September 2026 using one robot with seven joints per arm. The task prompt specified battery picking. Each action chunk contained 20 samples, with nominal planning velocity $2^{\circ}\mathrm{s}^{-1}$, a feedback hard limit of $4^{\circ}\mathrm{s}^{-1}$, and a pre-step stationary threshold of $0.25^{\circ}\mathrm{s}^{-1}$. Contact-control mode was disabled. These values describe the experimental configuration, not independently measured tracking performance.

Six execution sessions, R1–R6, provide configuration, environment, step, event, summary and driver records (Table 1). They contain 30 nested steps: 25 software-completed steps, four command-phase failures and one safety-phase rejection. Continuous bilateral telemetry is available for R2 step 5, R3 step 4 and R6 step 3, totaling six arm traces and 1,968 rows. The session inventory contains 11 entries, but five lack the artifacts required for this analysis. The six included sessions and three detailed steps form a convenience-selected experimental sample. They are not consecutive enrollment or independent fault replications.

R1–R5 used software development kit (SDK) 0.52.1 and configuration A; R6 used SDK 0.56.4 and configuration B. Configuration and version changed together. Differences among sessions therefore do not identify an SDK effect. Model identifier and training-step metadata were shared; model checkpoint and training data were unavailable for independent policy verification. Detailed source identity, missingness and duplicate handling are documented in Online Resource 1.

3.2 Signals, clocks and field prerequisites

Each arm-feedback row contains seven positions, velocities, efforts, currents, temperatures and external efforts. The primary kinematic vector is

$$\mathbf{x}_i = [\mathbf{q}_i^{\top}, \mathbf{v}_i^{\top}]^{\top}, \qquad \mathbf{q}_i, \mathbf{v}_i \in \mathbb{R}^7. \tag{1}$$

Execution and S1/S2 sampling data store position in radians and velocity in radians per second. Figures display degrees and degrees per second. Equality means equality at the saved numerical precision, not identity of the original transport bytes.

The acquisition chain provides HAL source time s_i, message-header time h_i and local reception time r_i, stored as integer nanoseconds. We define recorded age and reception interval by

$$a_i = (r_i - s_i)10^{-9}, \qquad \Delta t_i = (r_{i+1} - r_i)10^{-9}. \tag{2}$$

The saved age matches the first expression for every execution row. Monotonic clocks are not subtracted across sessions or reboots. A HAL timestamp is a source-layer clock, not a verified per-field acquisition timestamp. Sampling data also contain Data Distribution Service (DDS) source time; that clock remains distinct from header, HAL and reception time.

A contract $\mathcal{C} = (\mathcal{J}, \mathcal{F}, \mathcal{U}, \mathcal{T}, \mathcal{V})$ declares joint names, required fields, native units, one integration clock with continuity rules, and applicable producer-validity requirements. Four adapters map the measured formats to this shared representation. The validator matches unique names rather than inferring joints from array order. A position-only calculation does not require velocity, whereas a kinematic residual does. Missing or nonfinite values are retained as unavailable. A producer veto takes precedence over a finite numerical value; unspecified producer validity is not silently asserted true.

The reusable procedure is implemented by the four adapters and shared checks in Online Resource 2. It first resolves names and compatible arrays, then checks required fields and native units. It breaks the sequence at missing inputs, nonpositive intervals or excessive gaps. It returns numerical evidence and an evaluable predicate result, or an explicit reason for an undetermined

Table 1 Dual-arm execution sessions and feedback coverage. Completed steps denote software completion, not independently verified manipulation success. Raw telemetry covers only the named selected step; the remaining steps are represented by stored summaries and events

Session	SDK	Configuration	Session status	Steps	Completed	Detailed bilateral CSVs
R1	0.52.1	A	Failed	1	0	Unavailable
R2	0.52.1	A	Failed	5	4	Step 5
R3	0.52.1	A	Failed	4	3	Step 4
R4	0.52.1	A	Failed	1	0	Unavailable
R5	0.52.1	A	Stopped	16	16	Unavailable
R6	0.56.4	B	Failed	3	2	Step 3
Total				30	25	3 paired windows, 6 CSVs

result. Neither a finite array nor a not-refuted relation certifies physical measurement validity.

3.3 Position–velocity relation and error budgets

For an eligible window with reception-relative times t_i, the per-joint integral residual is

$$E_j = \sum_{i=1}^{n-1} \frac{v_{i+1,j} + v_{i,j}}{2}(t_{i+1} - t_i) - (q_{n,j} - q_{1,j}). \quad (3)$$

This compares displacement reported by position with displacement implied by the trapezoidal velocity sum. Both terms originate from feedback. The relation needs an allowance for quadrature, filtering and field alignment before it can support a physical interpretation.

Let $T = t_n - t_1$ and $a_{ij} = (v_{i+1,j} + v_{ij})/2$. Allow independent endpoint position errors ϵ_{qj}, per-sample velocity errors ϵ_{vj} and per-stamp time errors ϵ_t. A model allowance ϵ_{Ij} bounds discrepancy between latent endpoint displacement and the latent trapezoidal sum. The timing contribution has support bounded by $\epsilon_t S_j$, with

$$S_j = |a_{1j}| + \sum_{i=2}^{n-1} |a_{i-1,j} - a_{ij}| + |a_{n-1,j}|. \quad (4)$$

A necessary condition for the declared bounded relation is

$$|E_j| \le 2\epsilon_{qj} + \epsilon_{Ij} + \epsilon_t S_j + \epsilon_{vj}(T + 2\epsilon_t). \quad (5)$$

The required uniform velocity allowance therefore satisfies

$$\epsilon_{vj} \ge L_j = \frac{[|E_j| - 2\epsilon_{qj} - \epsilon_{Ij} - \epsilon_t S_j]_+}{T + 2\epsilon_t}, \quad (6)$$

where $[x]_+ = \max(0, x)$. With a fixed clock and independent box-bounded errors, L_j is the exact minimum. For uncertain timestamps it is a conservative lower bound. The common angular-velocity allowance is $\max_j L_j$; mixed native units remain separate. Online Resource 1 gives the proof and optimization formulation.

The implementation requires $\min_i \Delta t_i > 2\epsilon_t$, ensuring order for every allowed independent timestamp perturbation. Failure of this sufficient guard produces an undetermined result, not a proof that every ordered interpretation is impossible. An allowance below L_j refutes the stated bounded relation; otherwise the relation is not refuted. Missing units, velocity, required producer flags or finite error budgets prevent the corresponding calculation.

The full sensitivity grid uses endpoint and model allowances of $0, 0.001, 0.01, 0.1$ degrees and timestamp allowances of $0, 0.1, 1, 5$ ms. Velocity allowances span $0, 0.01, 0.1, 0.25, 1, 2\,^\circ\mathrm{s}^{-1}$. These are illustrative assumptions rather than calibrated specifications. The method reports their consequences without fitting a fault threshold.

3.4 Paired monitoring comparisons

The comparison evaluates recorded-age cutoffs, configured speed bounds, position-only small span, whole-message finiteness, required-arm finiteness and producer-aware field availability. Each pair uses the same measured rows or windows for the property under study. These are

transparent predicates, not reproductions of entire ROS packages or learned detectors. The age predicate uses the saved HAL-to-receiver clock pair; the speed comparator uses the configured $4^\circ\mathrm{s}^{-1}$ limit.

For a position-only rule,

$$B_q(\epsilon) = \mathbb{1}\left[\max_j\{\max_i q_{ij} - \min_i q_{ij}\} \le \epsilon\right]. \quad (7)$$

Exact kinematic residence is $H_i = (r_i - r_{b(i)})10^{-9}$, where $b(i)$ starts the current uninterrupted run of equal $\mathbf{x}$. A descriptive flag requires $H_i \ge \tau$ and $\max_j |v_{ij}| > 0.25^\circ\mathrm{s}^{-1}$. Its first occurrence is measured from the first observed sample, not an independently known fault onset. The proposed calculations were evaluated on the collected feedback; they were not deployed as a robot safety controller.

Every execution and sampling sequence is partitioned into nonoverlapping 0.5 s reception-time bins. A bin requires at least 0.4 s observed span, finite required fields, positive intervals and no gap above 0.1 s. Bins are anchored separately by block, topic and arm; no interpolation joins discontinuous segments. Gap limits of 0.05 and 0.5 s, age cutoffs of 10/20/50/100 ms and position tolerances of $0, 10^{-6}, 10^{-5}, 10^{-4}$ rad assess sensitivity. All eligible windows and all settings are retained.

3.5 Feedback-sampling experiment and supporting formats

The feedback-sampling experiment used a second seven-joint-per-arm robot on 13 October 2026. Sessions S1 and S2 used SDK 0.61.3; S2 followed a reboot. ROS best-effort subscriptions serialized group-state and aggregated `/joint_states` messages. These sessions contribute one robot-day, rather than two independent replications. Per-message acquisition identity and an external motion reference were unavailable.

The analysis sample consists of 16 nonoverlapping 4 MiB segments per session, at equally spaced byte offsets. The segment rule was specified after metadata/header characterization and before body sampling. It retained 134,217,728 bytes from 3,512,779,947 source bytes. Complete-line parsing yielded 90,451 records; 31 of 32 segments contained eligible arm messages. Equal byte spacing is not uniform time sampling, so high-volume phases can be overrepresented. Positions and velocities were selected by canonical joint names. Aggregated arm views share the feedback chain and are analyzed separately from direct group views.

Supporting schema evaluation uses 28 MCAP JointState records selected by storage/day/schema strata, retaining empty samples. Files up to 32 MiB contribute their complete content; larger files contribute three five-second intervals at 10%, 50% and 90% of duration. The full selection ledger appears in Online Resource 1. Storage groups are identifiers of data location, not verified independent robot platforms. The JointState definition permits empty optional arrays [18]; required values must therefore be assessed for the requested joint set. For arm names $\mathcal{J}_a$,

$$C_a(i) = \prod_{j \in \mathcal{J}_a} \mathbb{1}\{q_{ij}, v_{ij} \text{ are finite}\}. \quad (8)$$

The structural eligibility check is false for absent or duplicate required names and incompatible array lengths. Both-arm and all-joint predicates use the same message set. MCAP recording, publication and header times remain separate, since publication time can default to recording time and sequence is optional [19].

Three supporting bench records test producer-declared availability. Their data dictionary defines five rotary axes in radians, two actuator strokes in metres, and per-axis velocity-availability flags. Cross-storage duplicates are identified by decompressed hashes and counted once. These records test units and validity semantics; they do not add independent dual-arm fault cases.

3.6 Analysis timing, numerical verification and statistical scope

Monitoring comparisons and error-budget analyses were specified after measurement collection and initial signal inspection. The study is exploratory rather than preregistered or blinded. Selection rules, complete parameter grids and all paired outputs are supplied in Online Resource 2. No intentionally injected robot fault or additional physical trial was created for the numerical tests.

A separate linear program constructs sample-velocity corrections, endpoint corrections and

model discrepancy directly from each eligible window. It minimizes a common fixed-clock velocity allowance for four endpoint/model combinations per window. The reference uses SciPy 1.11.4 HiGHS and does not call the closed-form bound. Twenty-seven schema-invariant checks and twelve model-boundary checks assess adapters, missing inputs, budget monotonicity and order boundaries. Transformed test copies are software fixtures, not physical experiments. Analysis used Python 3.12.3, NumPy 1.26.4, pandas 2.3.3 and Matplotlib; complete scripts and dependencies accompany the data.

Frames, steps, bins and topic views are nested observations. The primary execution experiment is one robot-day; S1/S2 provide a second day under different task and SDK conditions. We report counts, ranges, empirical distributions and paired numerical contrasts. Population confidence intervals and hypothesis tests would not correct the selection and dependence. There is no independently labeled fault-onset or balanced healthy/fault set, so sensitivity, specificity and root-cause accuracy are not estimated.

4 Results

4.1 Execution outcomes define three monitoring conditions

The execution sessions exhibited distinct software paths. R1 and R4 failed on bilateral-goal acceptance with the right trajectory controller not running. R2 step 5 and R3 step 4 exceeded a reported velocity limit. R6 step 3 was rejected before command execution because feedback indicated motion, following a successful trajectory-result event. These observations distinguish controller readiness, excessive reported speed and pre-step admission.

The detailed feedback had finite position and velocity in all six arm traces (Table 2). HAL, header and reception timestamps increased within every trace. The right-elbow signals in Fig. 1 show the different velocity patterns. Continuous feedback for the other 27 steps is unavailable; their summaries and events do not reconstruct those trajectories.

4.2 Recent feedback can remain kinematically inconsistent

In R6 step 3, all 67 right-arm position–velocity vectors were identical across 0.663923 s. The median reception interval was 10.078 ms, and all 67 HAL and header stamps were distinct. Recorded age reached only 17.469 ms. Thus every row satisfied a 20 ms age rule while the kinematic payload remained fixed (Fig. 2).

The fixed right-elbow position was paired with velocity $1.897607^\circ\mathrm{s}^{-1}$. Over the full window, integrated velocity implied 1.259864° displacement while the reported position displacement was zero. The simultaneous left trace contained 68 distinct kinematic vectors and an elbow residual of -0.002133°. This contrast concerns feedback consistency; it does not establish which arm moved physically.

Field groups had different update patterns. All seven right-arm external-effort fields took two values, whereas position, velocity, effort, current and temperature stayed constant. The complete 42-field signal vector therefore repeated for at most 0.593470 s, shorter than the kinematic residence. The discrepancy would be obscured by describing the entire message as uniformly unchanged.

4.3 Sampling comparisons separate holding from missing velocity

The sampling data yielded 17,617 direct arm rows and 17,708 dependent aggregated views (Table 3). S1 direct rows contained complete velocity; S2 had 2,067 direct rows without velocity. No complete direct kinematic vector repeated in adjacent samples. The aggregated view had 95 equal adjacent pairs, lasting at most 12.770 ms. Neither view met the 0.10, 0.25 or 0.50 s nonzero-velocity residence rule.

Of 349 position-eligible direct bins, 311 had complete velocity and 38 lacked it. Position span in the latter ranged from 0.000824 to 17.652°. Replacing absent velocity with zero would impose a standstill interpretation on varying positions. These 38 bins therefore received undetermined kinematic outcomes rather than residuals. Among the 311 complete bins, 304 satisfied the 10^{-5} rad position-only small-span rule, illustrating why holding alone cannot identify the R6 discrepancy.

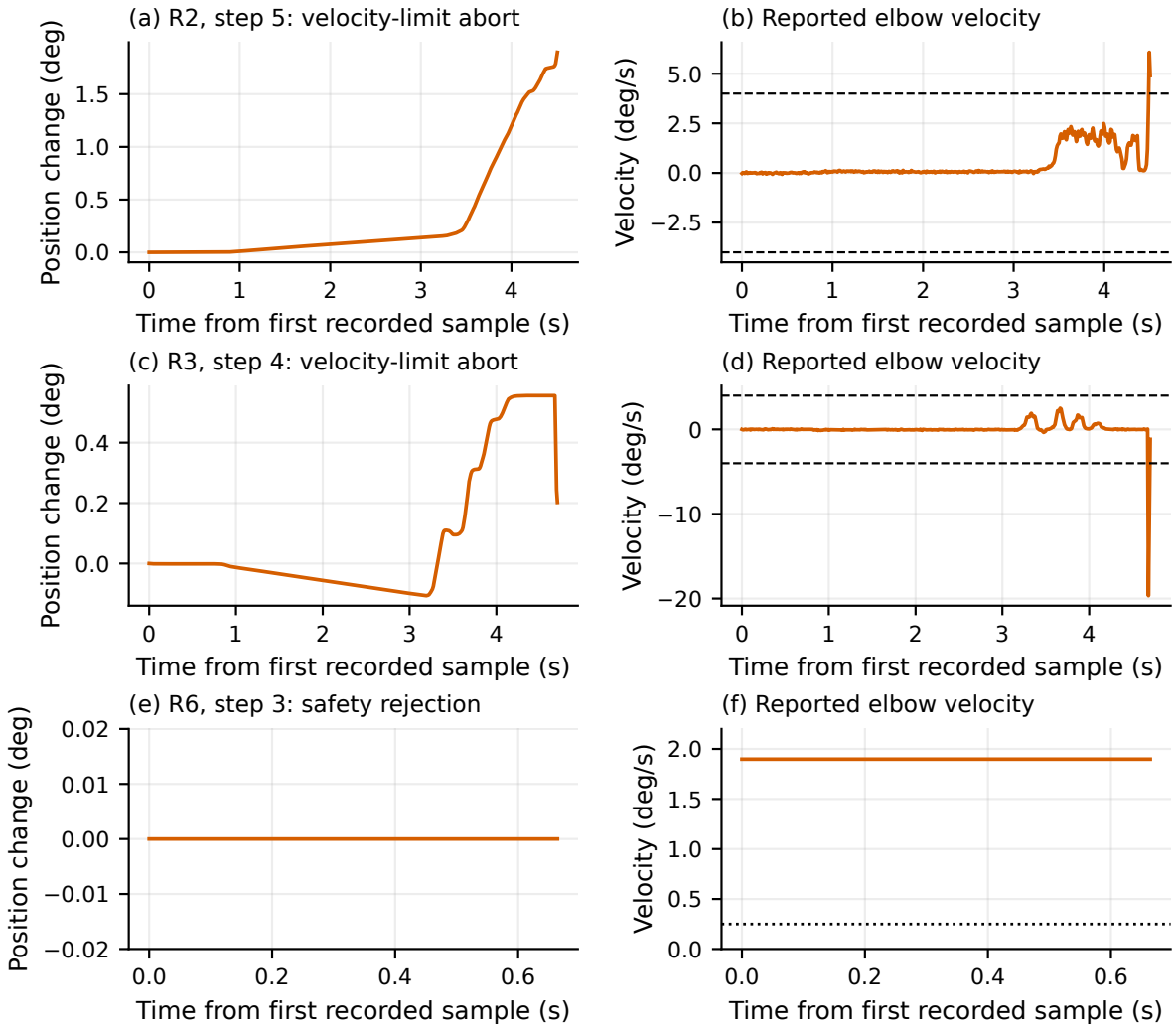


Fig. 1 Recorded right-elbow trajectories in the three selected steps. Panels (a), (c) and (e) show position relative to each window's first value; (b), (d) and (f) show reported velocity. Dashed horizontal lines in (b) and (d) indicate the configured $\pm 4°s^{-1}$ hard limit; the dotted line in (f) indicates the $0.25°s^{-1}$ stationary threshold.

Figure 3 summarizes these conditions. Changing the gap limit from 0.1 s to 0.05 or 0.5 s left the 311 direct and 302 aggregated finite-velocity bins unchanged. Aggregated views were excluded from the main between-session comparison because their source values largely matched the direct view (Online Resource 1).

4.4 Joint scope and producer flags change usable coverage

The supporting JointState data contained 77,627 messages, with finite position and velocity for both requested arms in every message. An all-joint finite-value check retained only 22,730 because 54,897 messages had nonfinite velocity outside those arms (Table 4; Fig. 4). Restricting the calculation to its declared arm set recovered 70.72% of the shared message denominator. It did not establish validity for the excluded joints.

Producer-aware checks had the opposite effect in the bench data. Six of 58,493 finite rows contained zero velocity with availability flags set to unavailable. Respecting those flags withheld six rows that a finite-only rule admitted. Together, these comparisons show that eligibility should

Table 2 Recomputed telemetry measures. Age is reception time minus the recorded HAL timestamp. Unique values refer to the complete 14-component position–velocity vector. E_{elbow} is the signed integral residual in Eq. 3; duration and residual use receiver time

Step	Arm	Rows	Duration (s)	Unique vectors	Maximum age (ms)	Peak $\lvert v\rvert$ ($^\circ s^{-1}$)	E_{elbow} ($^\circ$)
R2, step 5	Left	449	4.512	449	12.003	1.548	0.0053
	Right	448	4.508	448	12.169	6.082	−0.0015
R3, step 4	Left	468	4.703	468	10.785	2.094	−0.0020
	Right	468	4.703	468	13.360	19.664	0.0492
R6, step 3	Left	68	0.673	68	16.683	0.107	−0.0021
	Right	67	0.664	1	17.469	1.898	1.2599

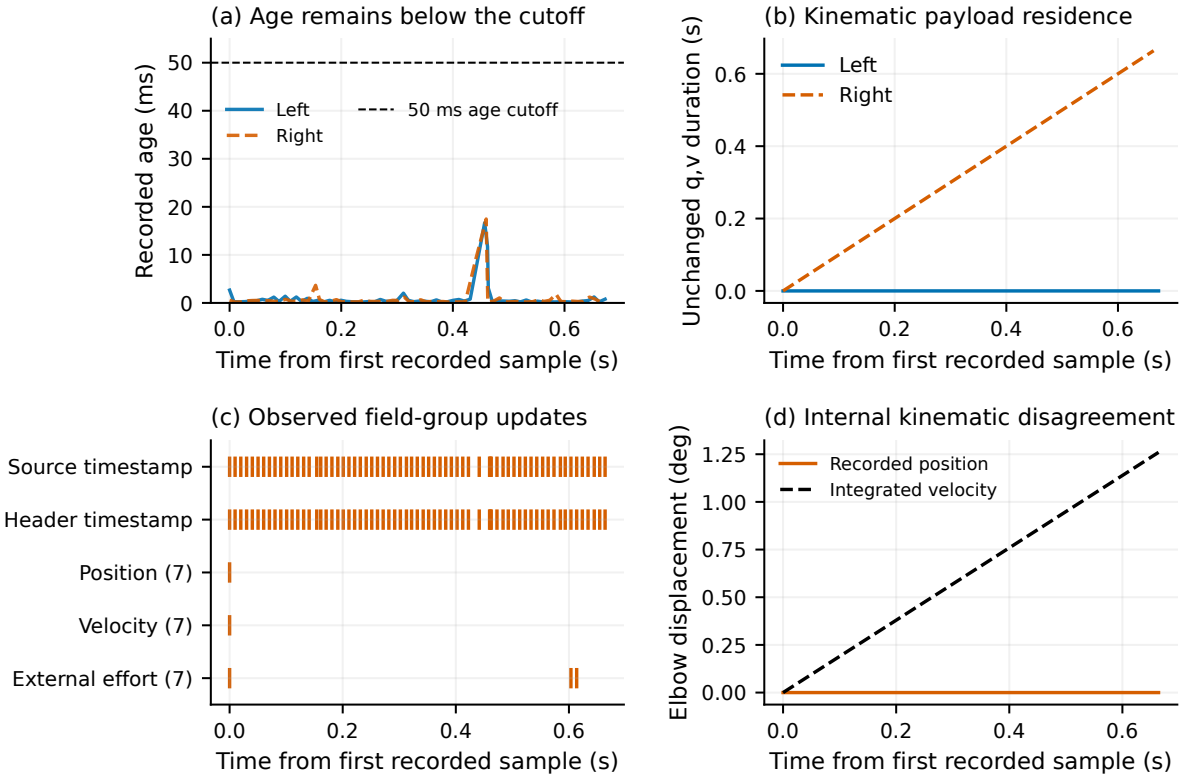


Fig. 2 Field-resolved evidence in R6 step 3. (a) Recorded age and a 50 ms descriptive cutoff. (b) Residence at an unchanged 14-component kinematic vector. (c) First values and observed numerical updates in the right-arm field groups; ticks represent changes, not verified acquisition events. (d) Elbow displacement obtained from position and from integrated velocity. The two arm windows use their own first received sample as time zero, so the display is a within-window comparison rather than sample-level bilateral synchronization

follow the requested property and applicable producer semantics, rather than maximize the number of accepted rows. All 27 schema-invariant checks passed across the four adapters.

Exact arm-vector residence in the JointState data reached 116.930 ms. The six runs exceeding 100 ms are numerical repetitions, not additional independently confirmed faults. All selected records, including empty samples, remain in the denominator ledger.

4.5 Paired monitoring rules have complementary blind spots

Table 5 compares predicates on identical inputs. The 20 ms age rule passed all 1,968 execution

Table 3 Feedback-sampling sessions, analyzed separately from the execution experiment. Rows are seven-joint arm views, not independent physical measurements. Equal pairs require finite position and velocity and a positive interarrival gap no greater than 0.1 s. Position and finite-velocity windows use the eligibility rules in Sect. 3.5

Session	Topic view	Arm rows	Missing velocity	Equal pairs	Max hold (ms)	Position windows	With velocity
S1	Direct group	7,370	0	0	0	144	144
S1	Aggregated	7,412	0	46	12.770	140	140
S2	Direct group	10,247	2,067	0	0	205	167
S2	Aggregated	10,296	2,068	49	12.709	200	162

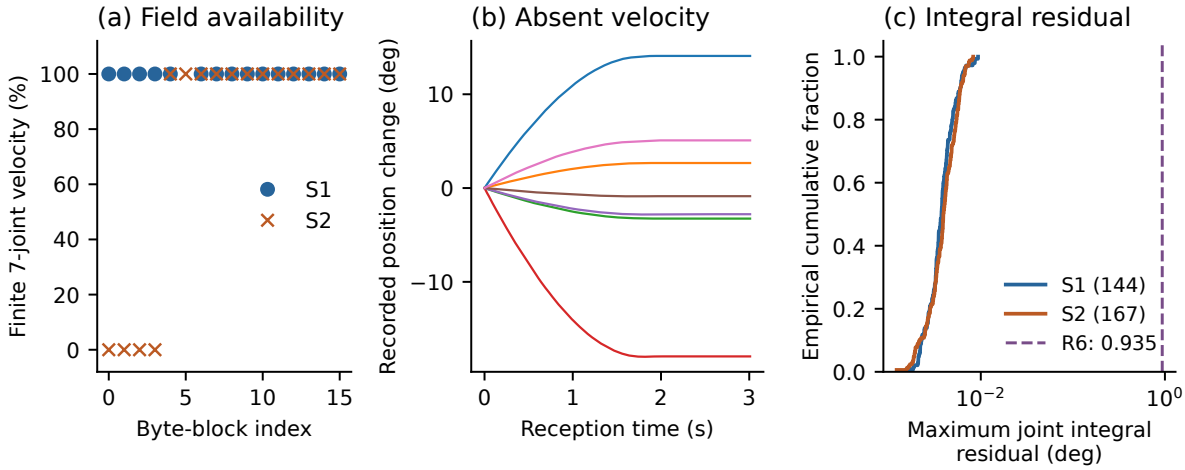


Fig. 3 Feedback-sampling comparison from nonoverlapping data segments. (a) Fraction of direct arm samples with finite seven-joint velocity; markers are not connected across blocks. S1 block 5 contains no eligible arm message. (b) All seven left-arm position changes in S2 block 1, the first nonzero-offset S2 block; every velocity array is empty. Colors distinguish joints. These are reported positions, not independent motion measurements. (c) Empirical distributions of maximum absolute joint integral residual in eligible direct windows (0.4–0.5 s observed span). The dashed line is the duration-matched R6 interval.

rows. The speed rule flagged two of the six arm traces, with right-elbow peaks of 6.081914 and 19.664209°s^{-1}. The held R6 trace remained below its 4°s^{-1} limit. Age and speed checks therefore did not expose its position–velocity disagreement.

The common-horizon comparison retained 38 execution bins and 311 direct sampling bins (Fig. 5). The held trace contributed one bin spanning 0.492742 s, with residual 0.935 degrees. The largest direct sampling residual was 0.00934 degrees. The matched horizon reduces duration as an explanation for that difference, while the missing-velocity bins remain explicitly outside residual calculation. This is a measured discrepancy contrast, not a calibrated decision threshold.

4.6 Error budgets quantify discrepancy and the abstention boundary

At zero endpoint and model allowances with a fixed clock, the selected held bin required 1.8976 degrees/s uniform velocity adjustment. The largest requirement among the 311 direct sampling bins was 0.01896 degrees/s, a 100.1-fold descriptive contrast. This adjustment measures the minimum correction to the discrete relation, rather than an estimated physical sensor error. All 1,396 independent linear programs agreed with the fixed-clock expression; the maximum implementation difference was below 3×10^{-14} degrees/s.

With 0.01-degree endpoint and model allowances and a 0.1 ms per-stamp allowance, the held bin still required at least 1.8352 degrees/s.

Table 4 Supporting JointState measurements by storage group. Counts retain every selected record, including empty numerical samples. Message counts concern the selected intervals or complete small recording. The two eligibility columns count messages: all named joints versus both seven-joint arms. Storage groups, records and arm views are not independent physical trials. A zero residence indicates no eligible adjacent equality, without establishing acquisition identity

Storage	Records selected/read	Messages	All joints finite q, v	Both arms finite q, v	Max arm residence (ms)
G1	2/2	1,500	0	1,500	0.000
G2	1/1	10,836	0	10,836	116.930
G3	21/21	42,561	0	42,561	0.000
G4	2/2	6,529	6,529	6,529	0.000
G5	2/2	16,201	16,201	16,201	0.000

Table 5 Paired monitoring-predicate comparisons on the same measured inputs. Each row states its own denominator and property. Counts describe outputs or eligibility, not correctly classified physical states. Different rows are not summed into a single performance score

Predicate	Observed output	Contract-based interpretation
Recorded age ≤ 20 ms	1,968/1,968 execution rows pass	Kinematic repetition and inconsistency remain separately evaluable
Configured speed bound	2/6 execution arm traces exceed $4^\circ s^{-1}$	Held R6 right remains below the bound; excessive speed and inconsistency differ
Position span $\leq 10^{-5}$ rad	304/311 finite-velocity direct sampling bins meet the rule	Small range alone does not supply a physical fault label
Required q, v for residual	311/349 direct sampling bins eligible	38 remain undetermined; no velocity imputation
Whole-message vs named arms	22,730 vs 77,627 eligible messages	54,897 additional arm-eligible messages; no claim for the unavailable joints
Finite vs producer-aware q, v	58,493 vs 58,487 bench rows eligible	Six unavailable zero-velocity rows withheld despite finite numbers

That lower bound exceeds the illustrative $0.25^\circ s^{-1}$ allowance. Under these assumptions, this was the only refuted execution bin; none of the 311 sampling bins was refuted. At 1 ms per stamp, however, the order guard withheld this bin. Its minimum interval was 1.384365 ms, placing the strict guard boundary at 0.692 ms (Fig. 6a).

Across the same position/model budgets and the $0.25^\circ s^{-1}$ velocity allowance, 1 ms timing uncertainty produced three undetermined execution bins and three undetermined sampling bins. The remaining 35 execution and 308 sampling bins were not refuted (Fig. 6b). At 5 ms, the guard withheld all 349 complete angular bins. Every setting remains in the supplied outputs. The practical output is therefore both a discrepancy budget and a stated condition under which the method cannot decide.

4.7 The contract separates observed events from causal conclusions

Table 6 specifies the evidence needed for each conclusion. The R6 successful trajectory-result event preceded the first held-window row by approximately 0.094 s. This orders the software events, but does not establish a physical stop between them. Likewise, cancellation handling after the R2/R3 exceptions documents a software response without measuring stopping time.

The contract retains these useful observations without promoting them to acquisition freshness or root cause. In particular, a repeated kinematic vector, advancing HAL stamps and changing auxiliary fields can coexist. None supplies the missing per-field device acquisition identity or external motion reference.

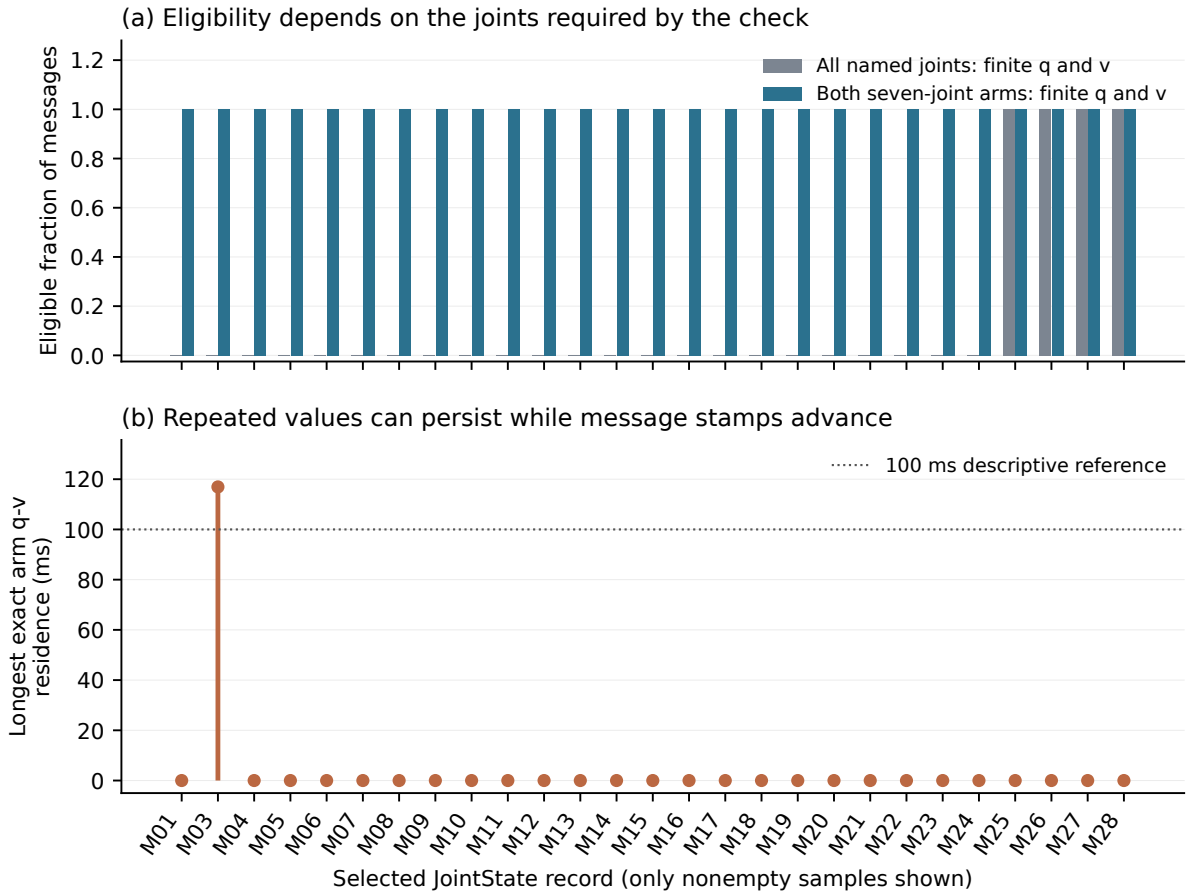


Fig. 4 The fields required by a diagnostic determine numerical eligibility. (a) Message-wide versus both-arm finite position–velocity checks in every selected record with decoded messages. The denominator is the same message set for both checks. (b) Longest exact seven-joint arm-vector residence per record, breaking at missing fields, nonpositive intervals and gaps above 0.1 s. The dotted 100 ms line is a descriptive reference. Header progression and numerical repetition describe the stored record; they do not supply physical fault labels. Unavailable records and empty measurement segments remain in Table 4 and the complete selection ledger

5 Discussion

5.1 Practical increment over individual checks

The method links three decisions that are often made separately: whether a requested signal is calculable, whether its position–velocity relation fits stated errors, and whether the evidence permits a physical interpretation. The experiments demonstrate why those decisions differ. A message can be recent yet inconsistent, globally incomplete yet usable for one arm, or numerically finite yet explicitly unavailable.

The gain over age and speed rules is a quantitative test of internal kinematic agreement. The gain over an unqualified residual is explicit dependence on units, field validity and error allowances. The gain over all-joint rejection is property-specific input retention. The timing guard supplies a further useful output: an uncertainty condition that withholds a verdict rather than quietly treating uncertain timing as exact. These gains concern the tested feedback properties; they do not replace a verified safety controller.

This composition extends observability-based diagnosis [2] by making the premises of a combined interpretation executable. It complements ROS diagnostics, tracing and temporal monitors [4, 5, 9]. Joint selection and three-valued outcomes are established ideas [10, 14]. The contribution is

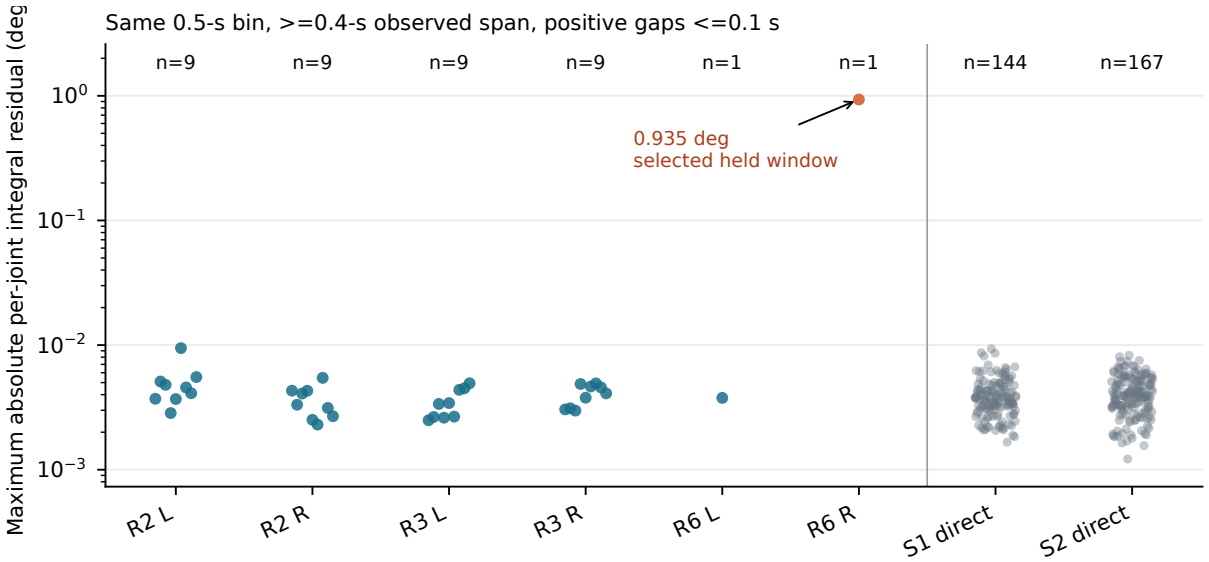


Fig. 5 Integral discrepancy at a common observation horizon. Every point is an eligible 0.5 s bin with at least 0.4 s observed span and positive reception gaps no larger than 0.1 s. The ordinate is the largest absolute joint residual in each bin, displayed in degrees. All 38 execution bins and 311 direct sampling bins are retained; the held right-arm trace contributes one eligible bin. Aggregated sampling views are excluded here because they depend on the direct view. Counts denote nested windows, not independent trials. Separation in this retained sample is descriptive and is not a calibrated physical-fault threshold

Table 6 Evidence contract for diagnostic statements. Undetermined denotes missing discriminating evidence, not an inferred normal condition. Evidence requirements are claim-specific rather than an ordinal quality score

Statement	Evidence required	Result for the measured inputs
Recorded timestamps progress	Ordered timestamps and positive differences	Established for the six raw arm windows
Required joint values are numerically present	Requested names, compatible array lengths and finite position/velocity values	Depends on the joint set; see Table 4. Producer flags and acquisition validity require separate checks
Kinematic payload is held	Finite fields, distinct vectors and residence	Sustained in R6; shorter repeats in aggregated sampling and MCAP records
Recorded kinematics disagree	Position increments and integrated velocity	Established internally in the held window
Execution was rejected or aborted	Phase-specific application and controller records	Established for the named software paths
A new device measurement was used	Original acquisition identity/time and field validity	Undetermined; HAL/message timestamps are insufficient
The robot moved or stopped physically	Synchronized independent motion observation	Undetermined
A particular cache or hardware fault caused it	Discriminating cross-layer command/feedback evidence	Undetermined

their connection to the bounded kinematic calculation, paired output comparisons and explicit uncertainty boundary. Compared with dynamic bounded-error observers [6], the simpler relation uses fewer system assumptions but cannot isolate a physical fault.

5.2 Interpretation and unresolved causes

Let $\mathcal{O} = \{s_i, h_i, r_i, \mathbf{q}_i, \mathbf{v}_i, \mathbf{z}_i\}$ denote measured clocks, kinematics, auxiliary fields and software events. A device or estimator can produce

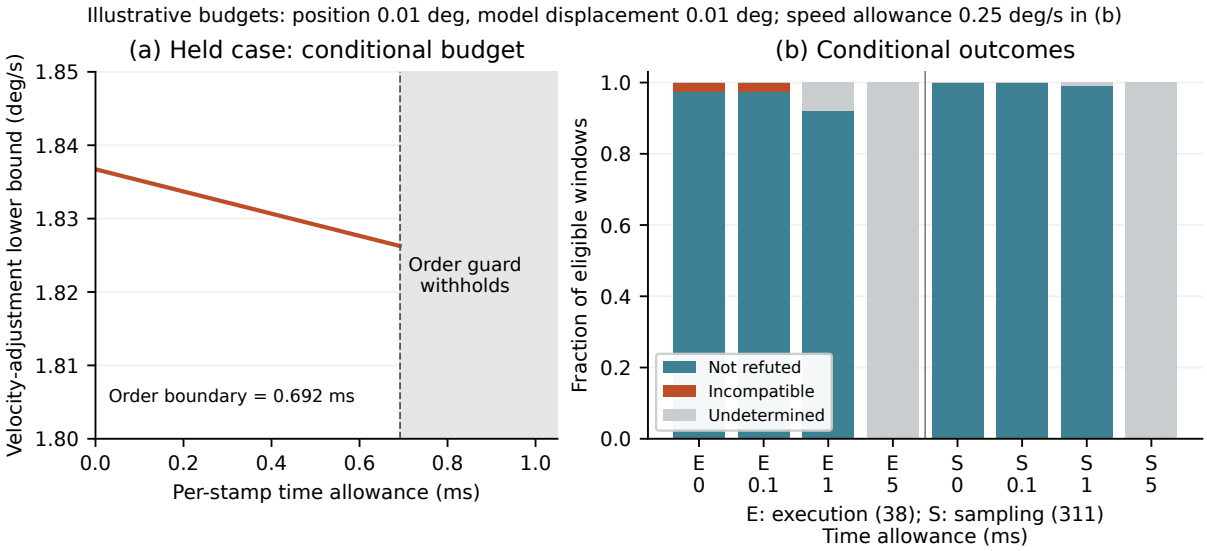


Fig. 6 Model-conditional budget analysis on the eligible measurement windows. (a) The selected held bin's velocity-adjustment lower bound with endpoint position and model-displacement allowances of 0.01 degrees each. Shading marks where the conservative stamp-order guard withholds a result. (b) Fractions of execution and direct-sampling bins that are incompatible, not refuted or undetermined under those same allowances and a velocity allowance of 0.25 degrees/s. Every timing setting is shown. The budgets are illustrative assumptions rather than calibrated physical error bounds; these outcomes are not fault-classification labels

repeated outputs through quantization, bias or an upstream defect. A downstream layer can instead reuse kinematics while assigning new message timestamps. Both mechanisms can generate the observed $\mathcal{O}$. A decision based on these signals alone therefore cannot select a unique causal layer.

The error budget narrows what can be said without resolving that ambiguity. A positive margin excludes the specified bounded relation. A nonpositive margin means only that this necessary condition does not exclude the relation. Unknown model discrepancy or failure of the timestamp guard prevents the proposed decision. This interpretation preserves the measured inconsistency while avoiding an unsupported physical fault label.

5.3 Monitoring design and experimental limits

A practical feedback interface should associate acquisition identity, acquisition time and validity with the field group they qualify. Publication time should remain separate. When that information is absent, downstream software can still report the calculated residual and budget, while keeping physical freshness undetermined. Controller completion, stationary admission and permission for the next action should use their own stated evidence requirements [16].

The sustained discrepancy remains one selected case from one execution robot-day. The second robot-day used different task and SDK conditions, so the contrasts do not identify a cross-robot or version effect. Only six of 11 execution entries and three detailed bilateral steps support the presented analysis. Byte-spaced segments and fixed MCAP intervals can miss temporal regimes. Four-adapter agreement establishes implementation reuse across formats, not independent hardware validation.

The error allowances were not physically calibrated. Filtering, time alignment and trapezoidal approximation must fit the model allowance before a physical interpretation is justified. The timestamp-order guard is conservative, and withholding need not imply that all ordered latent models are impossible. Shared feedback fields, missing device commands and the lack of synchronized external motion measurements limit root-cause and safety claims. These limits leave the computed within-window contrasts and eligibility comparisons intact.

The next discriminating experiment would synchronize per-field acquisition identifiers and final device commands with an external motion reference. Calibrated clock and field-error bounds would connect the conditional relation to measurement accuracy. Independent sessions with verified physical fault labels would then support estimates of detection performance.

6 Conclusion

Dual-arm execution and feedback sampling demonstrate the practical value of separating field eligibility from bounded position–velocity consistency. The contract exposes a discrepancy missed by age and speed checks, retains usable arm fields rejected by a global check, and honors producer-declared unavailability. Its minimum adjustment is exact for fixed clocks and verified by independent optimization. Explicit error allowances also identify when timestamp uncertainty requires abstention. The outcome is an executable procedure that returns an eligibility reason, a minimum adjustment budget and a conditional verdict. These results support monitoring under stated fields and assumptions; the single sustained case and uncalibrated budgets do not establish physical fault-detection accuracy or unique root cause.

Supplementary information

Online Resource 1 (`ESM_1.pdf`) supplies experimental coverage, sampling rules, schemas, sensitivity results and the bound proof. Online Resource 2 (`ESM_2.zip`) contains pseudonymized measurements, adapters, executable analysis, complete comparison outputs and figure code. Device/network identifiers and original private operational files are excluded.

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Data and code availability. Pseudonymized inputs, analysis code, parameters and results are included in Online Resource 2. Original operational files remain private because they contain infrastructure identifiers and access records.